

Reflective Lantern - Innovation Report

Project

Name: Ops-Vision

Date: 2026-08-26

URL: <https://github.com/atharvadevne123/reflective-lantern/tree/main/ops-vision>

Commits: 60 Files: 50 Tests: 192 passing Lint: ruff clean

What was built

FastAPI service exposing 10 endpoints: predict, predict/batch, incidents, health, metrics, runbooks/search, forecast, drift/status, plus root health and version probes.

Soft-voting ensemble - XGBoost + LightGBM + RandomForest - trained with 5-fold stratified CV scored on AUC-ROC.

sklearn Pipeline expanding 6 raw telemetry metrics into 10 engineered features, finished with RobustScaler.

KS-test drift monitor comparing a sliding production window against a reference distribution, flagging per-feature drift at $p < 0.05$.

FAISS runbook retrieval with a brute-force cosine fallback when faiss-cpu is absent, so a positive prediction returns a remediation runbook.

Holt double exponential smoothing forecasting 24h incident rate with 80% prediction intervals.

Airflow retraining DAG gated on 0.70 test AUC-ROC before promotion.

Docker (non-root, healthcheck) + compose with PostgreSQL 16, Alembic migrations, and GitHub Actions CI running lint, a 3.10/3.11 test matrix, mypy and a Docker build.

Bugs found and fixed during the build

1. The SQLAlchemy engine was constructed eagerly at import time, so simply importing app.database required a live psycopg2 driver. This broke test collection outright and would have broken CI. Fixed by making the engine lazy and dialect-aware, since SQLite rejects the QueuePool sizing options PostgreSQL needs.

2. The fitted feature pipeline was never persisted alongside the model. On any warm start - meaning any restart where a model file already existed on disk - the API loaded the saved model and paired it with a freshly built, UNFITTED scaler, so every prediction failed with HTTP 422. The fault is invisible to a single-pass test run because the first run trains in process and the pipeline happens to be fitted; it only appears on the second run. Fixed by loading model and pipeline as a matched pair, and a regression test now forces the reload path explicitly.

Environment constraints hit this run

Standalone repository creation returned HTTP 403. This session's GitHub scope is limited to atharvadevne123/reflective-lantern, so Ops-Vision was built inside that repository under ops-vision/ rather than as its own repo.

The three pre-flight sweeps (failing CI workflows, stray branches, missing releases) could not run: they enumerate all owned repositories, and that API path is outside this session's scope.

The Hacker News trending scrape was blocked by the network policy. The project topic was instead chosen from the SRE/observability domain and checked against all 25 previously built projects to avoid duplication.

SMTP is unreachable - outbound traffic is HTTPS-proxy-only, so ports 587 and 465 fail at connect. This report could not be emailed and is committed to the repository instead. Yesterday's run hit the same wall.

Status: 192/192 tests passing - ruff clean - 60 commits pushed to main